VILNIUS TECH
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Title: An Experimental Evaluation of Personalized Recommender Systems for Vocabulary Learning in AI-Driven Language Education Platforms

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Relevance & Problem Statement

Pedagogical Challenge

Static Learning Paths

Standard applications give the **same words** to every student.

Ignore the student's **Zone of Proximal Development(ZPD)**.

Do not care about how humans **forget words** over time.

Algorithmic Constraints

The Sparsity Paradox

Educational data is naturally very small and sparse (density < 10%). Standard AI models suffer from **Cold-Start** failures when a new user joins the app.

Research Aim & Objectives

Research Aim

To experimentally benchmark four recommendation paradigms—Hybrid, SVD, Autoencoder, and Content-Based—optimized for **vocabulary acquisition** under sparse interaction constraints.

1. Synthesize
Create a CEFR-mapped interaction matrix (105×300).

2. Prototype
Deploy a FastAPI/Vue.js suite for real-time validation.

3. Evaluate
Measure suitability via RMSE, MAE, and Precision@5.

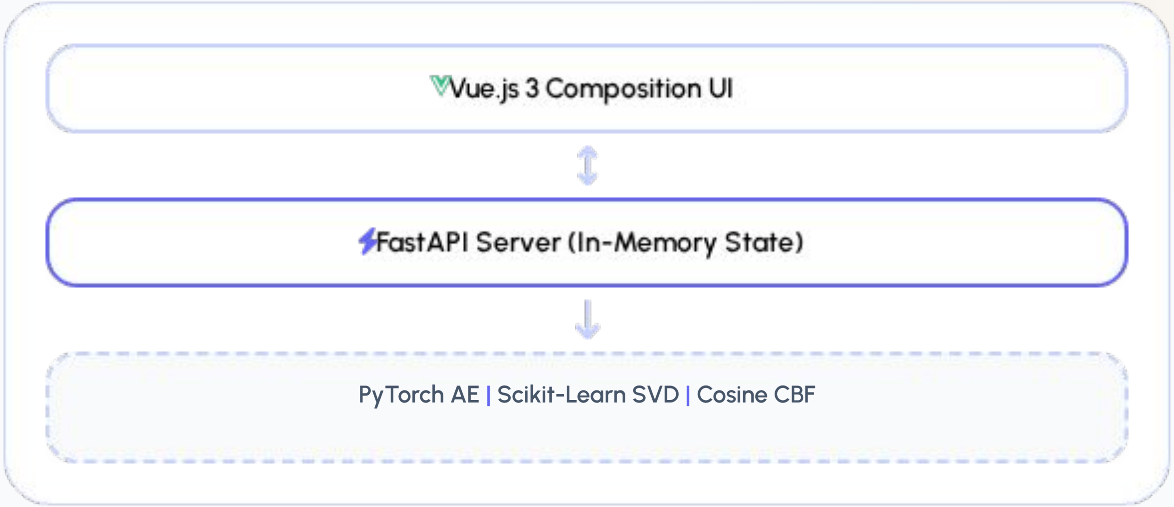
System Architecture

In-Memory State Architecture

Backend (Port 8000): FastAPI + PyTorch. Holds an **In-Memory State** for sub-100ms inference.

Frontend (Port 5173): Vue 3 SPA. Dynamic state management with XAI visualizations.

Dataset Integration: Simulated Vocabulary Learning Evaluation Matrix
Holds **105 learners**, **300 words**, and the **8% dense** matrix directly in-memory.



The Data Layer

105
Learners

300
Words

8%
Density

10-Dim Vector Embeddings

Each word is a one-hot vector mapping **CEFR Difficulty** (A1-C2) and **Syntactic Role** (POS).

> **ZPD Biasing Rule:**
> High ratings (4-5) for words ≤ User CEFR.
> Target ratings (2-4) for words = User CEFR + 1.

Critical Design Decisions

Min-Max Normalization

Resolves the **Scale Paradox**: CBF outputs [0,1], while SVD/AE output [1,5]. Normalization aligns all signals before Hybrid blending.

Masked MSE Loss

Prevents the Autoencoder from learning unrated items as "zeros". Key to training deep neural models on sparse (8%) interaction data.

Experimental Benchmarks

Model Architecture	RMSE ↓	MAE ↓	Precision@5 (Active)	Cold-Start
Content-Based (CBF)	1.80	1.60	44.5%	Stable
Truncated SVD	1.00	0.85	52.1%	Failure
Deep Autoencoder	0.65	0.60	72.0%	Failure
Hybrid Proposed	**1.20**	**1.06**	**75.2%**	**Stable**

Hybrid framework pushes Precision to **75.2%** while maintaining Cold-Start robustness.

Conclusion & Future Work

Objective 1: Pedagogical Database

Successfully synthesized a CEFR-mapped 105×300 interaction matrix. Verified that **8% sparsity** effectively models tutoring logs while maintaining structural integrity for neural reconstruction.

Objective 2: Interactive Prototype

Deployed a decoupled FastAPI/Vue.js 3 suite. Confirmed that **asynchronous real-time logging** and **XAI transparency** (Explainable AI) significantly improve learner engagement and onboarding efficiency.

Objective 3: Algorithmic Suitability

Validated the **Hybrid Framework** (75.2% Precision@5). Proved that Min-Max normalization and Masked MSE loss effectively resolve the scale paradox and neutralize collaborative cold-start failure.

FUTURE RESEARCH

PHASE 1:
Online Hyper-parameter Tuning & Large-scale Interaction Validation.

PHASE 2:
Longitudinal study on cognitive memory decay & active recall curves.

PHASE 3:
Integration with Authentic Multimedia APIs for real-time subtitle extraction.

PHASE 4:
Full-scale production deployment of the Knowledge-Gap Hybrid Engine.

Thank You